Output tables for 1xN statistical comparisons.

March 9, 2025

1 Average rankings of Friedman test

Average ranks obtained by each method in the Friedman test.

Algorithm	Ranking
Random	9.2
NN	9
Sweep	8.2
SaveSeq	2.8
SaveParall	1
MJ	3.4
CMT	4.8
Kilby	3.8
SaveMatch	6.2
FJ	6.6

Table 1: Average Rankings of the algorithms (Friedman)

Friedman statistic (distributed according to chi-square with 9 degrees of freedom): 38.323636.
P-value computed by Friedman Test: 0.000015.

2 Post hoc comparison (Friedman)

P-values obtained in by applying post hoc methods over the results of Friedman procedure.

i	algorithm	$z = (R_0 - R_i)/SE$	p	Holm	Finner	Li
9	Random	4.28231	0.000018	0.005556	0.005683	0.034357
8	NN	4.177864	0.000029	0.00625	0.011334	0.034357
7	Sweep	3.760077	0.00017	0.007143	0.016952	0.034357
6	FJ	2.924505	0.00345	0.008333	0.022539	0.034357
5	SaveMatch	2.715611	0.006615	0.01	0.028094	0.034357
4	CMT	1.984485	0.047202	0.0125	0.033617	0.034357
3	Kilby	1.462252	0.143672	0.016667	0.039109	0.034357
2	MJ	1.253359	0.210075	0.025	0.04457	0.034357
1	SaveSeq	0.940019	0.347208	0.05	0.05	0.05

Table 2: Post Hoc comparison Table for $\alpha = 0.05$ (FRIEDMAN)

Holm's procedure rejects those hypotheses that have an unadjusted p-value ≤ 0.0125.
Finner's procedure rejects those hypotheses that have an unadjusted p-value ≤ 0.033617.
Li's procedure rejects those hypotheses that have an unadjusted p-value ≤ 0.034357.

3 Adjusted P-Values (Friedman)

Adjusted P-values obtained through the application of the post hoc methods (Friedman).

i	algorithm	unadjusted p	p_{Holm}
1	Random	0.000018	0.000166
2	NN	0.000029	0.000235
3	Sweep	0.00017	0.001189
4	FJ	0.00345	0.0207
5	SaveMatch	0.006615	0.033077
6	CMT	0.047202	0.188807
7	Kilby	0.143672	0.431016
8	MJ	0.210075	0.431016
9	SaveSeq	0.347208	0.431016

Table 3: Adjusted p-values (FRIEDMAN) (I)

i	algorithm	unadjusted p	p_{Finner}	p_{Li}
1	Random	0.000018	0.000166	0.000028
2	NN	0.000029	0.000166	0.000045
3	Sweep	0.00017	0.000509	0.00026
4	FJ	0.00345	0.007746	0.005257
5	SaveMatch	0.006615	0.011876	0.010032
6	CMT	0.047202	0.06996	0.067432
7	Kilby	0.143672	0.180792	0.180387
8	MJ	0.210075	0.23302	0.243462
9	SaveSeq	0.347208	0.347208	0.347208

Table 4: Adjusted p-values (FRIEDMAN) (II)